

Nafisa Islam Rifa

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 [Nafisa Islam Rifa](#) |  [NafisaIslamRifa](#) | [Portfolio](#) | [Google Scholar](#)

London , United Kingdom

EDUCATION

- **University Of Greenwich** September 2025 - October 2026 (Expected)
MSc in Data Science London, United Kingdom
 - Current Average: 71% (Distinction), 120/180 credits completed
- **United International University** February 2022 - Aug 2025
Bachelor of Computer Science and Engineering Dhaka, Bangladesh
 - GPA: 3.95/4.00
- **Rajshahi College** May 2019
Higher Secondary Certificate Rajshahi, Bangladesh
 - GPA: 5.00/5.00
- **Govt PN Girls' High School** March 2017
Secondary School Certificate Rajshahi, Bangladesh
 - GPA: 5.00/5.00

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

- [C.1] (Accepted) *Iftekhar Mahmud, Puja Rani Das, Nafisa Islam Rifa, Irfan Hossain, Raiyan Rahman and Md. Farid Dewan, Multimodal Emotion Recognition Using Visual and Thermal Image Fusion: A Deep Learning Approach*, IEEE ICCIT 2024 [IEEE Conference].
 - Worked on multimodal fusion of visual and thermal images
 - Performed dataset preprocessing and model experimentation
- [C.2] (Accepted) *Nafisa Islam Rifa, Visual Grounding and Explainability for Prompt-Driven Radiology Report Generation*, 7th International Conference on Medical Imaging and Computer-Aided Diagnosis (MICAD 2026).
 - Developed a parameter-efficient radiology vision-language model combining frozen Rad-DINO, a spatial adapter, and a 4-bit LoRA-adapted Qwen2.5-3B decoder.
 - Introduced explicit coordinate-token supervision for jointly generating radiology reports and anatomically grounded bounding boxes.
 - Developed an evaluation framework to audit visual grounding and explainability by comparing generated localizations with annotated regions and CAM- and prior-based baselines.
- [J.1] (Under Review) *Nafisa Islam Rifa and Mohammad Nomani Kabir, Video Moment Retrieval: A Survey of Methods, Benchmarks, and Open Challenges in the Multimodal LLM Era*, Computer Vision and Image Understanding (Elsevier), (Q2, IF: 3.6).
 - Surveyed 165 papers (2017–2026) and developed a unified taxonomy covering pre-MLLM, Transformer/VLP, and MLLM-era methods across five dimensions.
 - Audited 34 MLLM-based methods, identifying gaps in bias-corrected evaluation, Video Corpus Moment Retrieval, and spatio-temporal grounding.
 - Presented cross-era performance comparisons on major VMR benchmarks, along with consolidated results for VCMR and spatio-temporal grounding.

PROJECTS

- **Build Large Language Model from Scratch** 2025 
 - Developed a GPT-like Large Language Model from scratch based on the book *Build a Large Language Model (From Scratch)* by Sebastian Raschka.
- **Agentic AI-Powered London Tube Assistant using RAG and Live TfL APIs** Jul 2026 – Present 
 - Developed an intelligent London Tube assistant using **Retrieval-Augmented Generation (RAG)**, **agentic AI routing**, and the **Transport for London (TfL) API** to answer static and real-time transport queries.
 - Built a RAG pipeline with **LangChain**, **ChromaDB**, **sentence-transformers**, a **cross-encoder reranker**, and a local **Llama 3.2 (Ollama)** model for grounded response generation.
- **Network Intrusion Detection by Lightweight Deep Learning Models** 2025 
 - Built a network intrusion detection system using the augmented UNSW-NB15 dataset with feature engineering for robust performance.
 - Compared Logistic Regression, Decision Tree, Random Forest, SVM, and XGBoost models, with Random Forest achieving the highest accuracy.

SKILLS

- **Programming Languages:** Python, R, SQL
- **Data Science & Machine Learning:** scikit-learn, PyTorch, TensorFlow, XGBoost, Regression, Classification, Clustering, Time-Series Analysis, Feature Engineering, PCA
- **Data Analysis & Wrangling:** Pandas, NumPy, Excel (Advanced)
- **Databases:** MySQL, MongoDB, Cassandra
- **Data Visualization:** Matplotlib, Seaborn, Plotly, Power BI, Tableau
- **Tools & Technologies:** Git, Roboflow, Figma

SERVICES

- Reviewer, **MI4MedFM Workshop at MICCAI 2026**
- Reviewer, **WiML Workshop @ NeurIPS 2026**

LANGUAGE PROFICIENCY

- **English:** Proficient, IELTS Academic – Overall Score: 7.5 (Listening: 8.0, Reading: 7.5, Writing: 6.5, Speaking: 7.5)
- **Bengali:** Native Language

REFERENCES

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